The speed of sound in air at 20°C is 344 m/s. Use this speed to calculate the wavelength in cm of a 737-Hz sound wave.

λ = _____cm

Suppose a wave has frequency 25.0 Hz and wavelength 3.30 m. Answer the following questions:

(a) The speed of the wave is

(b) The angular frequency is rad/s.

(c) The period is

(d) The wave number is rad/m.

A wave equation is given by $\alpha \frac{\partial^2 y}{\partial t^2} - \beta \frac{\partial^2 y}{\partial x^2} = 0$, where $\alpha = 3$ s^2 and $\beta = 258.6$ m^2. Calculate the wave speed.

v = _____m/s

Suppose a wave function is given by $y(t,x) = 2\cos(4t - 2.2x + 1.4)$, with all values in MKS units. Then ...

(a) the amplitude is [Select] m;

(b) the angular frequency is [Select] rad/s;

(c) the wave number is [Select] rad/m;

(d) the phase shift is [Select] rad;

(e) the frequency is [Select] Hz;

(f) the period is [Select] s;

(g) the wavelength is [Select] m;

(h) the propagation speed of the wave is [Select] m/s;

(i) the transverse **speed** of the wave at (0,0) is [Select] m/s.

A wire of mass 13 g and length 1.3 m under a tension of 711 N has an amplitude of 0.85 cm. If the frequency is 152 Hz, calculate the average power in watts of a wave propagating on the wire.

P_{av} = _____ W

Two traveling waves of the same frequency (f = 200 Hz) and wavelength interfere to produce a standing wave. The distance from node #1 to node #5 is 4.02 m. What is the wavelength (in m) of the original waves?

A vibrating string has a fifth harmonic of 1,436 Hz. What is the second harmonic in Hertz?

A wire with fundamental frequency 339 Hz and length 0.37 m is under tension of 1,472 N. Calculate the wire's mass per unit length in **grams/meter**.

$\mu =$ _____ g/m (Note the units!)

A wire with mass 0.051 kg and length 0.67 m has a fundamental frequency of 140 Hz. Calculate the tension in the wire in Newtons.

A given wave will have a certain number of wavelengths per meter. The number of radians corresponding to that number of wavelengths is called the [blank].

Using only words from the list below, place the correct or best word in each blank in the following sentences concerning standing waves on a string fixed at both ends. Words may be used more than once.

The distance from a node to an adjacent antinode is

of a wavelength. The distance from one end of the string to the other is

of a wavelength at the fundamental frequency.

Word list: **one-eighth; one-quarter; three-eighths; one-third; one-half; five-eighths; two-thirds; three-quarters**

A wave whereby the vibration is perpendicular to the direction of propagation is called a [Select] wave, whereas a wave having the vibration parallel to the direction of motion is called a [Select] wave.

The fact that the linear combination of any number of solutions of the wave equation is also a solution to the wave equation is called the principle of [blank].